

# Choi Hei Wang (Oscar)

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## EDUCATION

### The Chinese University of Hong Kong

Hong Kong

*B.Sc. in Computer Science; Minors: Data Analytics and Informatics, Business*

Aug 2023 – Expected Jul 2027

- Relevant courses: Compiler Construction (A), Operating Systems, Algorithms, Data Structures, Computer Organization, Probability, Statistics, Linear Algebra; current: Quantitative Methods for Financial Derivatives, Quantum Computer Systems (postgraduate)
- Final-year project (Prof. XU Hong): built a decision layer that makes small local LLM agents safer; on held-out tasks it cut rule violations by 80% versus a prompted agent (Bayesian scoring + conformal risk control)

## EXPERIENCE

### CUHK Quant Trading Society

Jan 2026 – Present

*Lead Developer, Quant Research Team*

Hong Kong

- As the team's sole developer, architected its market-data platform, holding 700B+ rows of Chinese A-share tick data (2017–2026), crypto market data and Polymarket order books
- Designed its query service for researchers (Python, HTTP, SQL): a full stock-day loads in under 0.2 s, and a test with 12 readers, 4 appending writers and a concurrent partition rewrite lost no data
- Led data cleaning: removed 500M+ duplicate rows and blocked look-ahead leaks into research data
- Run the team's shared GPU research server: per-member permissions (PAM/eBPF modules in C), 40+ live health probes, and a Go relay that moved ~300 Mbit/s of market data off the VPN, ending remote-login drops

### WorldQuant BRAIN

Jun 2025 – Present

*Research Consultant, Gold level*

Remote

- Researched and submitted 11 equity alphas (best out-of-sample Sharpe 2.18); built Python tooling for the BRAIN API that manages simulation queues under concurrency limits with checkpoint/resume

## COMPETITIONS

### IMC Prosperity 4 (Team SHDC)

2026

*Top 5% globally (904th / 18,803 teams); 18th in Hong Kong*

- Team of 3: coded the final two rounds and checked the team's results; market making with Kalman/OU fair values plus relative-value trading

### Hong Kong Universities Web3 Quant Trading Hackathon

2025

*Initiated the project; lead architect*

### MonsoonSIM Enterprise Resource Management Competition

2025

*3rd place*

## PROJECTS

*Self-directed; designed and verified by me, implemented with AI coding agents (Claude Code, Codex).*

### AI Quant Research System | Python, Claude Code, C++20

2026

- Orchestrated about 140 AI subagents through 266 gated tasks to build a tested Python library from a quant textbook: 1,400+ claims verified by passing tests, with notes in a versioned research vault
- Rewrote the order-book replay engine in C++20: 25× faster than optimized Python with byte-identical output

### GPU Backend for FatQat (open-source quantum simulator) | Python, CUDA (CuPy)

2026

- Added CUDA (GPU) execution for all four simulation methods of an open-source quantum-circuit simulator, keeping its interfaces and double precision: 17–21× faster than its own optimized 32-thread CPU engine on 24–28-qubit observables, within 2.9 machine epsilons of a 60-digit reference, or 23–26× in an optional plain-arithmetic mode; fix merged upstream

### Formulaic Alpha Search (GP) | Python, genetic programming

2026

- Genetic programming over formulaic alpha expressions on 6.7 years of public Binance data (34 coins), with pre-registered train/validation/test splits, delay-1 signals, trading costs and an equal-budget random-search control

## TECHNICAL SKILLS

**Languages:** Python, C++, C, SQL, Java

**Data & Systems:** DuckDB, Apache Arrow/Parquet, SQLite, Linux, Docker, Git

**Quant, ML & AI:** NumPy, pandas, PyTorch, Qlib, Claude Code, Codex; factor research, market microstructure

**Spoken:** Cantonese (native), English (fluent), Mandarin (fluent)